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(54) **TENSOR COMPRESSION**

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CPC ***G06N 3/063*** (2013.01); ***G06F 7/22*** (2013.01); ***G06F 7/50*** (2013.01); ***G06F 7/523*** (2013.01); ***H03M 7/3059*** (2013.01); ***H03M 7/6064*** (2013.01)

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(57) **ABSTRACT**

Lossy tensor compression and decompression circuits compress and decompress tensor elements based on the values of neighboring tensor elements. The lossy compression circuit scales each decompressed tensor element of a tile by a scaling factor that is based on the maximum value that can be represented by the number of bits used to represent a compressed tensor element, and the greatest value and least value of the tensor elements of the tile. The lossy decompression circuit performs the inverse of the lossy compression. The compression circuit and decompression circuit have parallel multiplier circuits and parallel adder circuits to perform the lossy compression and lossy decompression, respectively.

20 Claims, 7 Drawing Sheets

